Project Title

Resume–Job Matching & Talent Marketplace Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by
Mallipeddi Harshika (2520090047)
Chilumula Madhulika (2520090056)

Team Number: 18

Section: 10

Under the guidance of

Mrs. Anitha Chekuri

Assistant Professor, CS-IT

Hyderabad-500090, Telangana, India.



K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



DECLARATION

The Mini Project Report entitled **"Resume–Job Matching & Talent Marketplace Engine "** is a record of Bonafide work of **Mallipeddi Harshika - 2520090047, Chilumula Madhulika - 2520090056** submitted in partial fulfillment for the award of B. Tech in Computer Science and Information Technology to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

Mallipeddi Harshika - 2520090047

Chilumula Madhulika - 2520090056

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



CERTIFICATE

This is to certify that the mini project based report entitled "**Resume–Job Matching & Talent Marketplace Engine ”** is a bonafide work done and submitted by **Mallipeddi Harshika - 2520090047, Chilumula Madhulika - 2520090056** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science and Information Technology, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my guide, Mrs. Anitha Chekuri, Assistant Professor, CS&IT which cannot be expressed in words for her tremendous support, encouragement and guidance for this project.

We express our gratitude to **Dr. S. Khaja Shareef (CS&IT)**, Head of the *Department for Computer Science and Information Technology* for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

Mallipeddi Harshika - 2520090047

Chilumula Madhulika - 2520090056

TABLE OF CONTENTS

S.No.	Contents	Page no.
1.	Abstract	6
2.	Introduction	7-8
3.	Literature survey	9-11
4.	Problem Statement and Objectives	12-13
5.	Hardware and Software requirements	14
6.	Implementation	15-18
7.	Results & Discussion	19-23
8.	Conclusion & Future Scope	24-25
9.	References	26

1.ABSTRACT

The Résumé–Job Matching & Talent Marketplace Engine is a recruitment-oriented platform developed to simplify the process of connecting students with suitable job opportunities. In the current job market, students often face difficulty in identifying roles that match their skills and qualifications, while recruiters spend considerable time screening and comparing multiple résumés. The proposed system addresses these challenges by providing separate modules for students and recruiters. Students can create profiles, upload résumés, and obtain extracted information such as skills, education, and projects, while recruiters can post job requirements and search for suitable candidates.

The core of the project is a matching engine that applies Data Structures and Algorithms such as hashing, string matching, arrays, and priority queues to compare candidate skills with job requirements and generate compatibility scores. Candidates are then ranked according to their matching scores, enabling recruiters to identify suitable applicants efficiently. The system also provides skill gap analysis, career roadmap suggestions, and mock interview preparation to support students in improving their employability.

The project demonstrates how DSA concepts can be applied to a practical recruitment scenario, improving candidate selection efficiency while helping students understand and address the skills required for their desired career opportunities.

Team Name: 18

Team Members:

1. Mallipeddi Harshika - 2520090047
2. Chilumula Madhulika - 2520090056

2. INTRODUCTION

The recruitment process plays an important role in connecting students and job seekers with suitable career opportunities. With the increasing number of graduates and job openings, both candidates and recruiters face difficulties during the recruitment process. Students may find it challenging to identify jobs that match their skills, educational qualifications, and project experience. At the same time, recruiters need to examine a large number of résumés and identify candidates who best satisfy the requirements of a particular job. Traditional résumé screening methods can be time-consuming and may not always provide an efficient way to compare candidates.

The **Résumé–Job Matching & Talent Marketplace Engine** is developed to address these challenges by providing a platform that connects students and recruiters through an algorithm-based matching approach. Students can create their profiles, upload résumés, and provide information about their skills, education, and projects. Recruiters can post job requirements by specifying the skills and qualifications expected from candidates. The system processes this information and compares candidate profiles with available job requirements.

Data Structures and Algorithms form the core of the project. Techniques such as hashing, string matching, arrays, and priority queues are used to efficiently process, compare, and rank candidate information. A compatibility score is generated based on the similarity between candidate skills and job requirements, allowing suitable candidates to be prioritized. The system also includes features such as skill gap analysis, career roadmap generation, and mock interview preparation, which help students improve their employability.

In addition to simplifying the recruitment process, the proposed system aims to improve transparency and provide useful insights to both candidates and recruiters. Candidates can understand how closely their skills align with specific job requirements and identify areas where they need improvement. Recruiters can use the matching results and candidate rankings to narrow down the list of potential applicants and focus on those who are more relevant to the

position. By organizing candidate information and job requirements in a structured manner, the platform reduces the effort involved in manual screening and supports more informed recruitment decisions. These features make the system useful as both a recruitment support tool and a career development platform for students and job seekers.

The main objective of this project is to demonstrate the practical application of Data Structures and Algorithms in solving a real-world recruitment problem. By combining algorithmic matching with career-support features, the project aims to make candidate selection more efficient and help students identify opportunities that align with their skills and career goals.

The remainder of this documentation presents the problem statement, objectives, existing and proposed approaches, system requirements, system design, Data Structures and Algorithms used, implementation, results, testing, and conclusion of the project.

3. LITERATURE SURVEY

3.1 RésuMatcher – A Personalized Résumé–Job Matching System

RésuMatcher is an automated system developed to match résumés with suitable job opportunities. The system extracts important information from résumés and compares it with job requirements to identify relevant positions. It uses techniques such as skill extraction, pattern matching, and similarity measurement for matching candidates with jobs. This project demonstrates how automated résumé processing can reduce the effort involved in manually screening candidates. It provides a useful foundation for developing efficient résumé–job matching applications.

3.2 Semantic Matching of Job Seekers to Vacancies

This project focuses on matching job seekers with suitable vacancies using semantic matching techniques. Candidate information and job requirements are analyzed to determine the relevance between a person and a particular vacancy. The approach considers skills and qualifications rather than depending only on exact keyword matches. It demonstrates how automated comparison can improve the quality of candidate-job recommendations. This concept is relevant to our project because our matching engine also compares candidate skills with job requirements.

3.3 AI-Based Resume and Job Description Matching

This system focuses on automatically comparing résumés with job descriptions to determine candidate suitability. It applies natural language processing and semantic similarity techniques to identify relationships between candidate profiles and job requirements. The system generates matching scores that can help recruiters shortlist suitable candidates. This reduces the time required for manual résumé screening. The approach provides a foundation for implementing automated candidate ranking in recruitment applications.

3.4 Intelligent Resume Analysis and Job Matching

This project combines résumé analysis with automated job matching to support recruitment activities. Important information such as candidate skills, education, and experience is extracted from résumés and compared with job descriptions. The system uses automated analysis to recommend suitable job opportunities to candidates. It also demonstrates how skill gaps can be identified by comparing existing candidate skills with required skills. These concepts are closely related to the skill-gap analysis and matching features of our project.

3.5 AI-Driven Job Matching Platform

An AI-driven job matching platform uses candidate profiles and job descriptions to generate suitable job recommendations. The system analyzes multiple candidate attributes and calculates the relevance of available opportunities. Automated ranking helps recruiters identify candidates who best satisfy the requirements of a particular position. Such platforms reduce the time and effort involved in traditional recruitment. The study highlights the importance of efficient matching and ranking techniques in modern recruitment systems.

3.6 Knowledge Graph-Based Job Matching

Knowledge graph-based recruitment systems represent relationships between candidates, skills, occupations, and job requirements. These relationships can be used to identify suitable candidates even when the terminology used in a résumé differs from the terminology in a job description. The approach provides more structured information for candidate-job matching. It can also support explainable recommendations by showing the relationships between skills and job requirements. This provides useful ideas for improving the accuracy and transparency of our matching engine.

3.7 Resume Screening and Candidate Ranking System

Automated resume screening systems are designed to reduce the workload of recruiters by analyzing large numbers of candidate résumés. Such systems extract relevant candidate information and compare it against predefined job requirements. Candidates can then be ranked according to their level of compatibility with the position. This approach improves the

efficiency of the initial recruitment stage. Our project follows a similar concept while emphasizing the practical use of Data Structures and Algorithms for matching and ranking.

3.8 Online Talent Marketplace Platforms

Online talent marketplace platforms connect candidates with organizations based on skills, qualifications, and job requirements. Candidates create profiles containing their professional information, while recruiters publish available positions and search for suitable applicants. Matching and recommendation features help reduce the time required to discover relevant opportunities. These platforms demonstrate the practical importance of connecting candidate profiles with employer requirements. Our project extends this concept by incorporating DSA-based matching, skill-gap analysis, career roadmaps, and mock interview preparation.

3.9 Research Gap

Existing résumé and job-matching systems mainly focus on automated résumé analysis, semantic similarity, artificial intelligence, and recommendation techniques. However, the Résumé–Job Matching & Talent Marketplace Engine focuses on demonstrating the practical application of Data Structures and Algorithms in a recruitment environment. The proposed system uses hashing, string matching, arrays, and priority queues for skill comparison, compatibility scoring, and candidate ranking. In addition, it combines recruitment functionality with skill-gap analysis, career roadmap generation, and mock interview preparation, providing benefits to both students and recruiters.

3. Problem Statement and Objectives

Problem Statement:

To develop a résumé–job matching platform that efficiently compares candidate skills and qualifications with recruiter-defined job requirements using Data Structures and Algorithms.

Problem Description:

Students often find it difficult to identify job opportunities that match their skills, education, and project experience, while recruiters need to screen a large number of résumés to find suitable candidates. The proposed system addresses this challenge by extracting candidate information, comparing it with job requirements, generating compatibility scores, and ranking suitable candidates.

Objectives:

The main objectives of the Résumé–Job Matching & Talent Marketplace Engine are:

1. To develop a student profile module that allows students to create profiles containing their education, skills, projects, and other relevant information.

2. To provide résumé upload and information extraction so that important details such as skills, education, and projects can be obtained from candidate résumés.

3. To develop a recruiter module that allows recruiters to post job requirements, including required skills and qualifications.

4. To apply Data Structures and Algorithms such as hashing, string matching, arrays, and priority queues for efficient candidate-job matching.

5. To calculate compatibility scores by comparing candidate skills with the skills required for available job positions.

6. To rank suitable candidates according to their compatibility scores so that recruiters can identify relevant candidates more efficiently.

7. To identify skill gaps by comparing the skills possessed by a candidate with the skills required for a desired job.

8. To provide career roadmap and mock interview features that help students improve their skills and prepare for suitable career opportunities.

9. To demonstrate the practical application of Data Structures and Algorithms in solving a real-world recruitment problem.

4. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 10 GB of available storage
- **System:** Desktop computer or laptop
- **Internet Connection:** Required for accessing the application and related resources

Software Requirements:

- **Operating System:** Windows 10/11 or Linux
- **Programming Languages:** Java, HTML, CSS, JavaScript
- **Development Environment:** Visual Studio Code
- **Database:** MySQL
- **Backend:** Java
- **Frontend:** HTML, CSS, JavaScript
- **Web Browser:** Google Chrome or Microsoft Edge
- **Version Control:** Git and GitHub
- **Additional Tools/Libraries:** Apache Tika for résumé text extraction and required Java libraries for implementing the matching algorithms.

5. IMPLEMENTATION

The implementation of the Résumé–Job Matching & Talent Marketplace Engine is divided into multiple modules. Each module performs a specific function and together they provide the complete recruitment and career-support functionality of the project. The major modules implemented in the project are Student Profile and Résumé Processing, Recruiter Job Posting, Job Matching and Candidate Ranking, and Skill Gap Analysis.

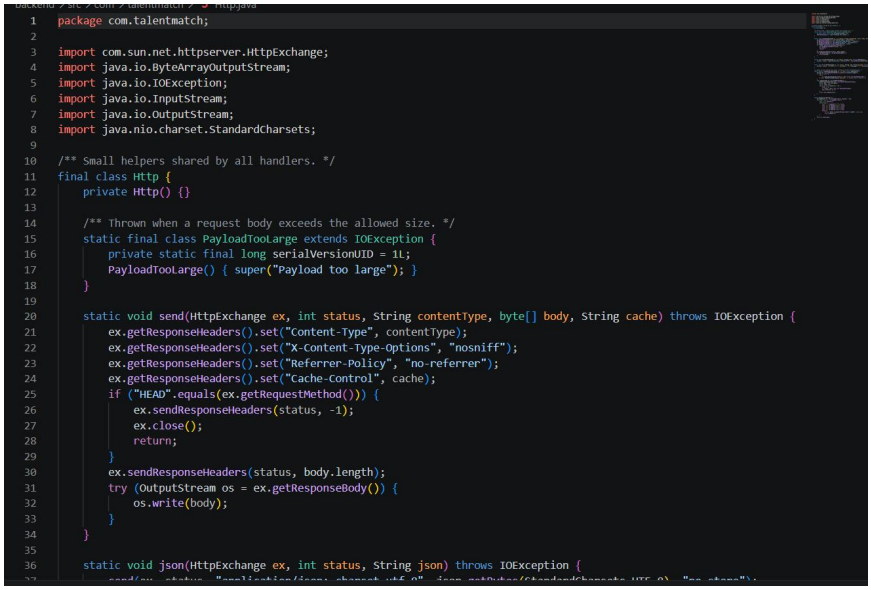
6.1 Student Profile and Résumé Processing

This module allows students to create their profiles and upload their résumés. The résumé is processed to extract important information such as name, email, education, skills, and projects. The extracted information is stored and used by the matching engine for comparing the candidate with available job requirements.

Technologies/Concepts Used:

- Java
- Résumé text extraction
- Arrays
- String processing

Sample Code:



```java
package com.talentmatch;

import com.sun.net.httpserver.HttpExchange;
import java.io.ByteArrayOutputStream;
import java.io.IOException;
import java.io.InputStream;
import java.io.OutputStream;
import java.nio.charset.StandardCharsets;

/** Small helpers shared by all handlers. */
final class Http {
    private Http() {}

    /** Thrown when a request body exceeds the allowed size. */
    static final class PayloadTooLarge extends IOException {
        private static final long serialVersionUID = 1L;
        PayloadTooLarge() { super("Payload too large"); }
    }

    static void send(HttpExchange ex, int status, String contentType, byte[] body, String cache) throws IOException {
        ex.getResponseHeaders().set("Content-Type", contentType);
        ex.getResponseHeaders().set("X-Content-Type-Options", "nosniff");
        ex.getResponseHeaders().set("Referrer-Policy", "no-referrer");
        ex.getResponseHeaders().set("Cache-Control", cache);
        if ("HEAD".equals(ex.getRequestMethod())) {
            ex.sendResponseHeaders(status, -1);
            ex.close();
            return;
        }
        ex.sendResponseHeaders(status, body.length);
        try (OutputStream os = ex.getResponseBody()) {
            os.write(body);
        }
    }

    static void json(HttpExchange ex, int status, String json) throws IOException {
```

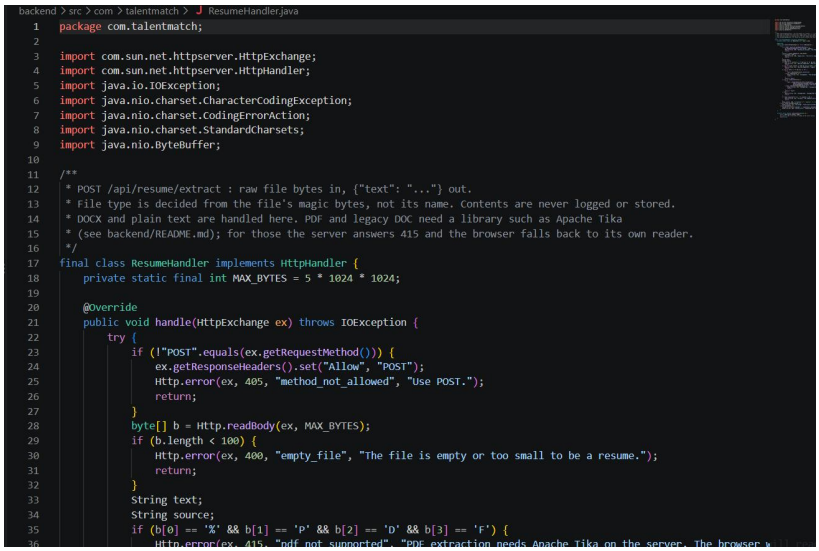
6.2 Recruiter Job Posting

This module allows recruiters to create job postings by specifying the job title, required skills, and other qualifications. The required skills are stored and passed to the matching engine. These requirements are then compared with the skills available in student profiles.

Technologies/Concepts Used:

- Java
- Arrays
- String processing
- Data storage

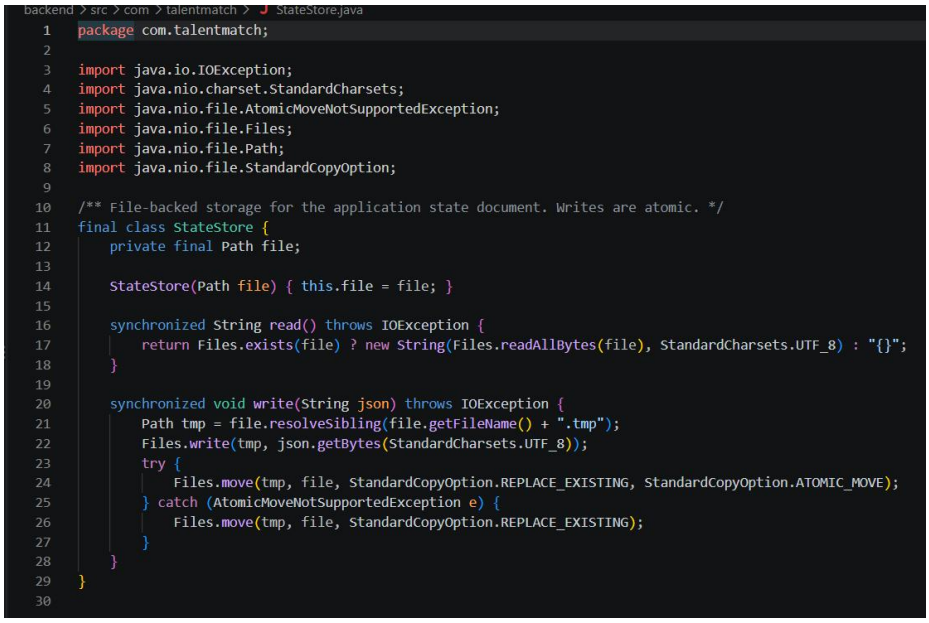
Sample Code:



6.3 Job Matching and Candidate Ranking

This is the core module of the project. The candidate's skills are compared with the skills required for a particular job. String matching and hashing can be used to identify matching skills efficiently. A compatibility score is calculated based on the number of matching skills. The candidates can then be organized using a priority queue so that candidates with higher compatibility scores receive higher priority.

Sample Code:



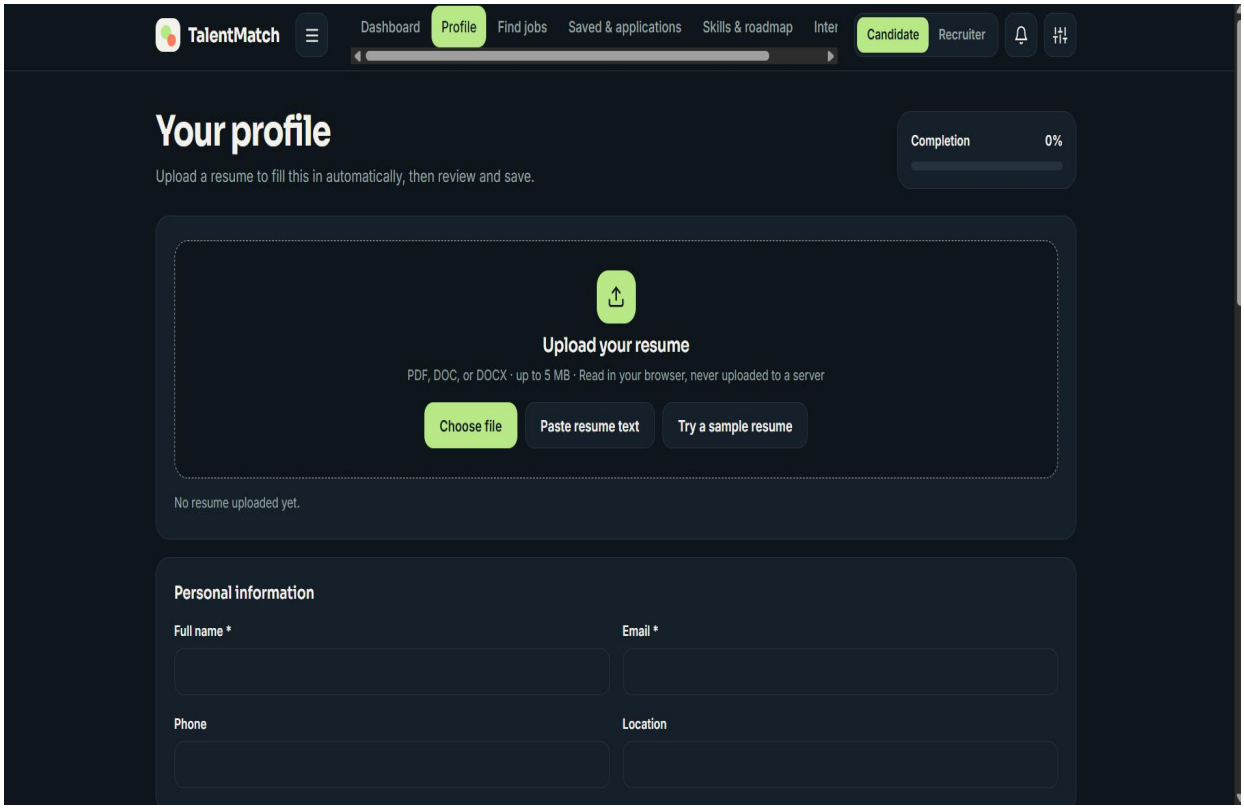
6. RESULTS & DISCUSSION

The Résumé–Job Matching & Talent Marketplace Engine was implemented to demonstrate the practical application of Data Structures and Algorithms in the recruitment process. The system provides functionality for student profile creation, résumé processing, recruiter job posting, candidate matching, candidate ranking, and skill-gap identification. The implemented modules were tested using sample candidate profiles and job requirements.

7.1 Student Profile and Résumé Processing

The student module successfully accepts candidate information and résumé details such as education, skills, and projects. The extracted information is used as input for the matching process.

Output Explanation:
The student profile is created successfully, and relevant résumé information is displayed in a structured format. The extracted skills and qualifications are subsequently used for job matching.

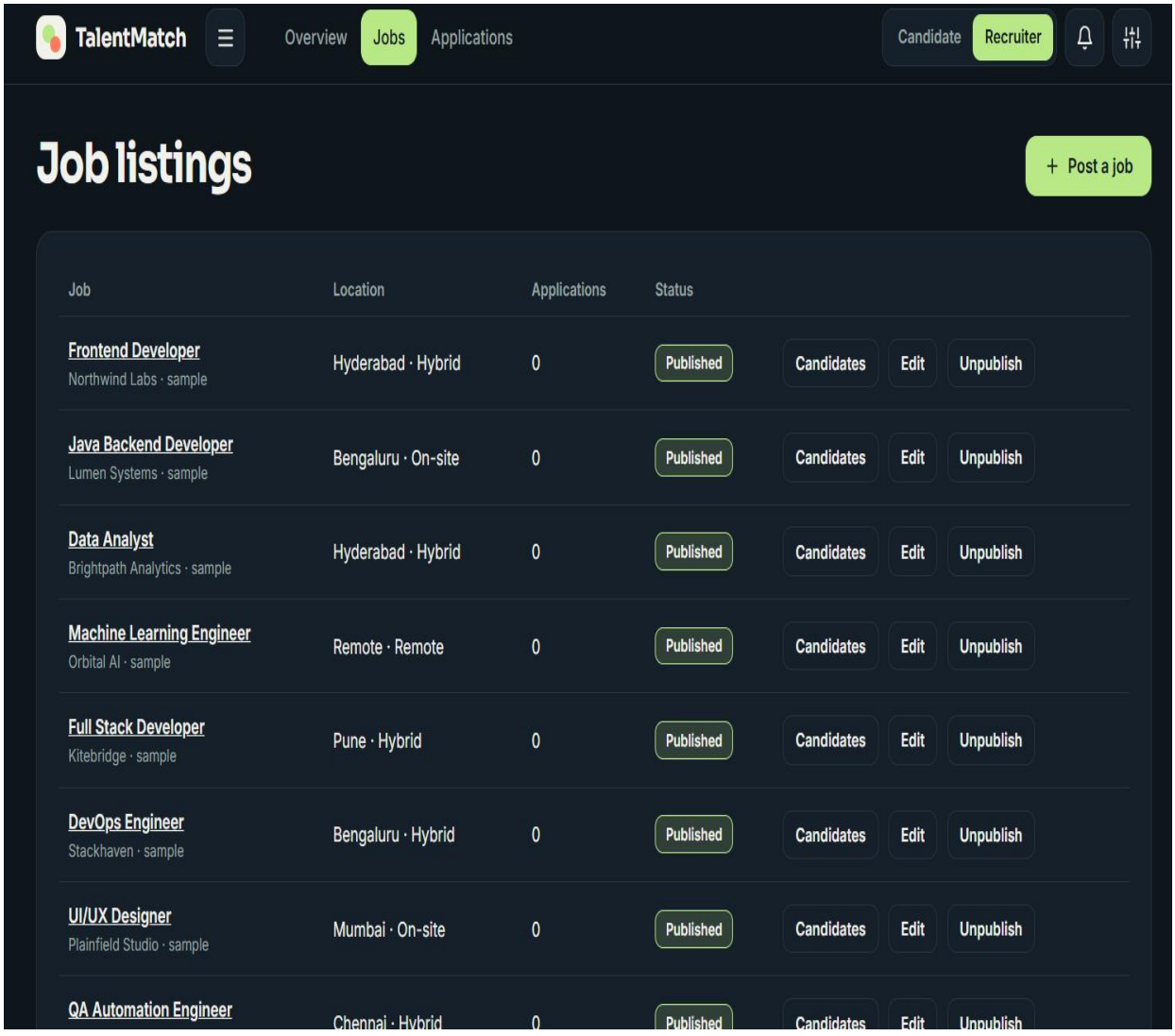


7.2 Recruiter Job Posting

The recruiter module allows recruiters to enter job details and specify the skills required for a particular position. These requirements are stored and used by the matching engine.

Output Explanation:

The job posting is created successfully with the required skills. The system uses these requirements to compare and evaluate available candidates.

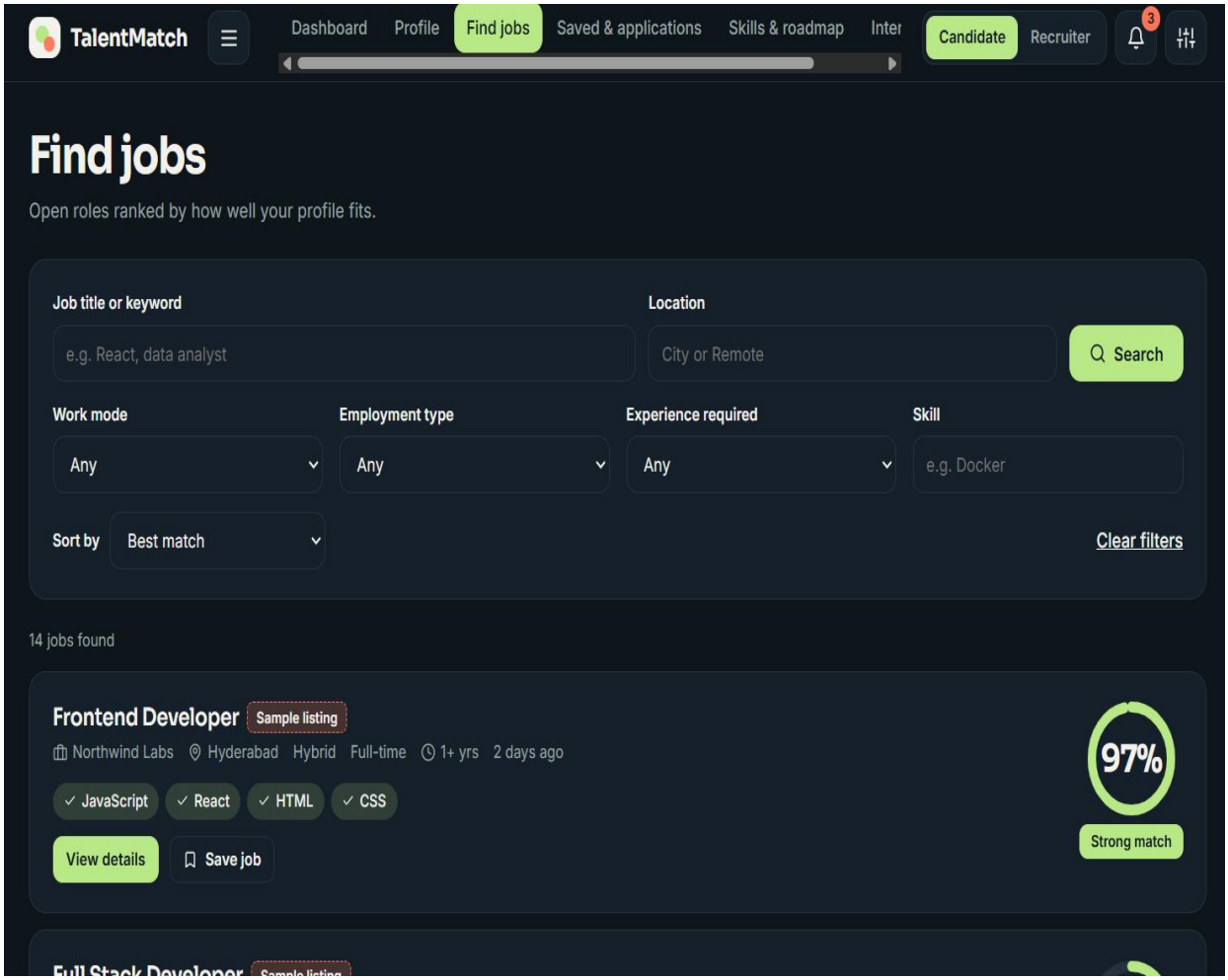


7.3 Job Matching and Compatibility Score

The matching engine compares candidate skills with the required skills of a job. Hashing and string matching techniques are used to identify matching skills, and a compatibility score is calculated for each candidate.

Output Explanation:

The system successfully calculates a compatibility score based on the number of matching skills. Candidates with higher compatibility scores are identified as more suitable for the specified job.

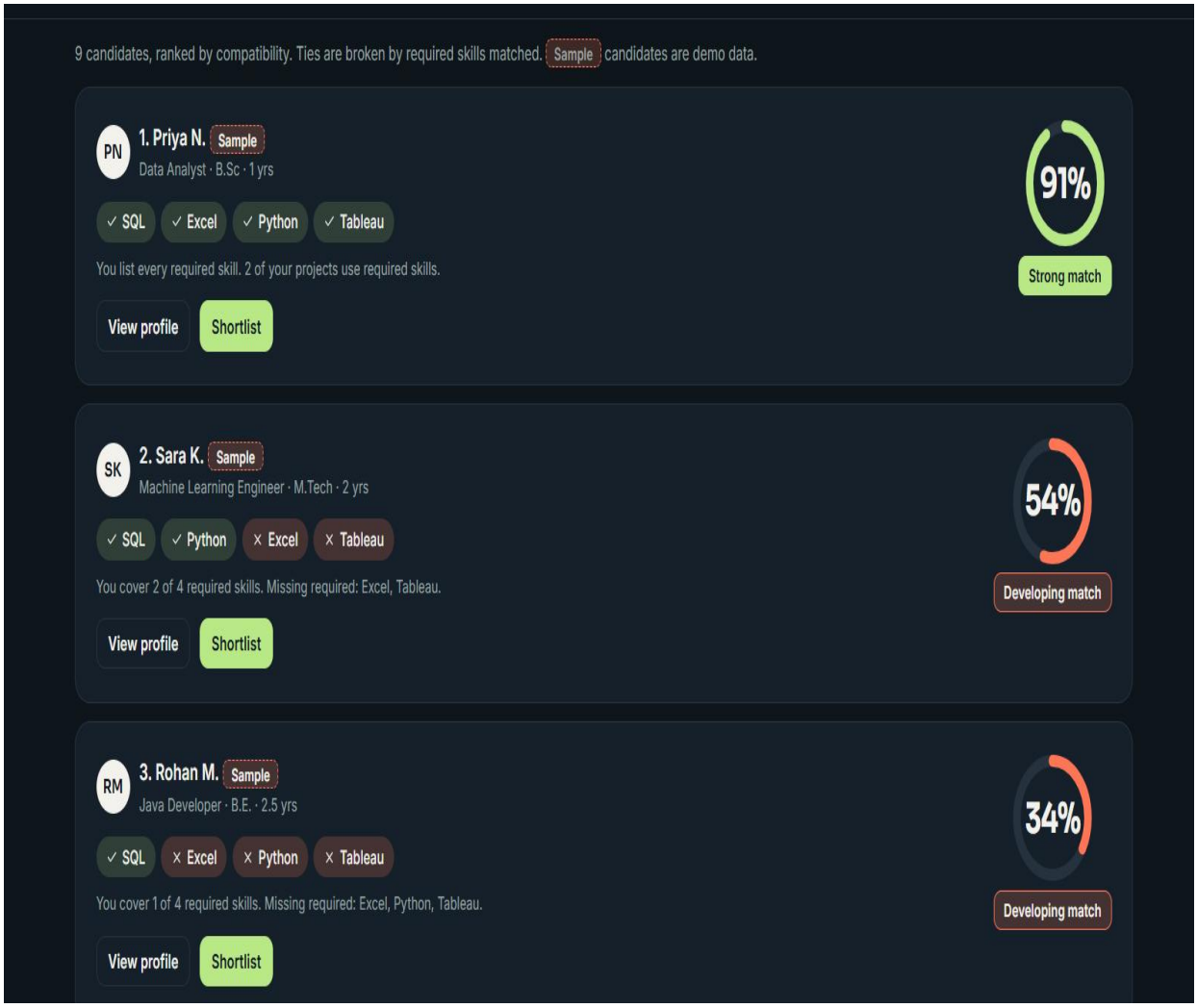


7.4 Candidate Ranking

The matching results are used to rank candidates according to their compatibility scores. A priority queue can be used to give higher priority to candidates with better matching scores.

Output Explanation:

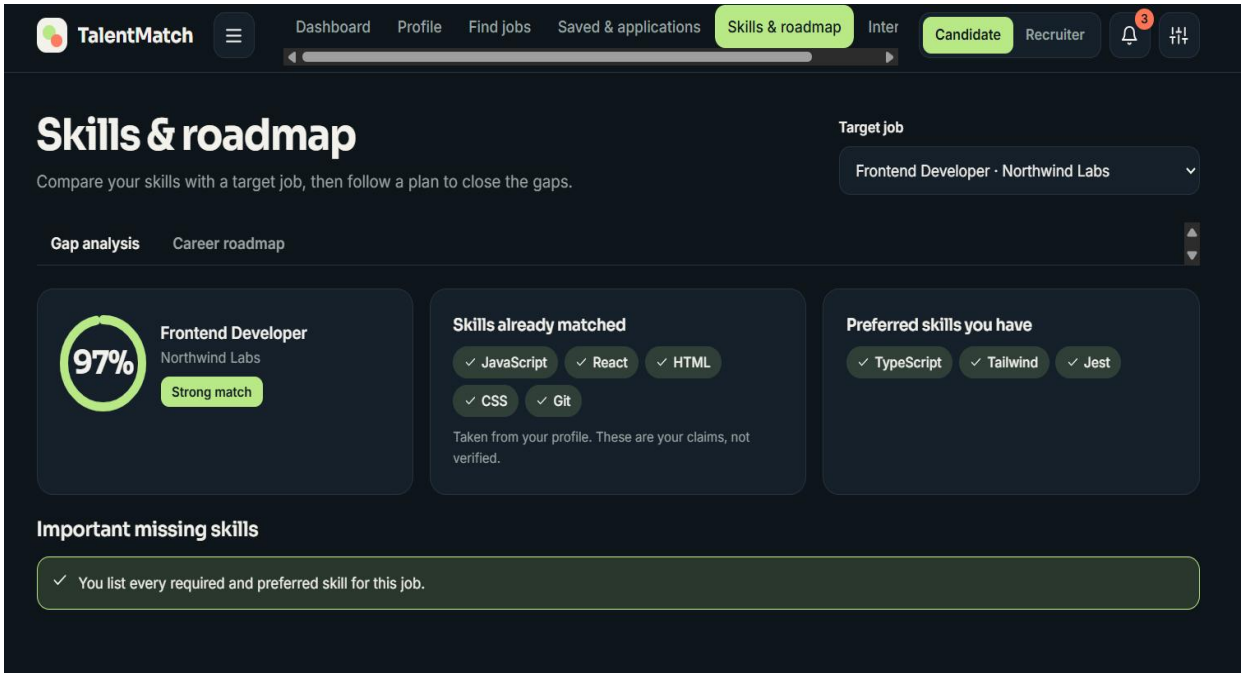
Candidates are arranged according to their compatibility with the job requirements. This helps recruiters identify suitable candidates more efficiently.



7.5 Skill Gap Analysis

The skill-gap module compares the skills possessed by a candidate with the skills required for a selected job. Skills that are required but missing from the candidate profile are identified.

Output Explanation:
The system displays the missing skills for the selected job role. This helps students understand the areas they need to improve to become eligible for their desired opportunities.



7.6 Overall Discussion

The results demonstrate that the proposed system can apply Data Structures and Algorithms to a practical recruitment problem. Hashing enables efficient skill lookup, string matching supports skill comparison, arrays store candidate and job information, and priority queues assist in candidate ranking. The combination of these techniques provides an efficient approach for comparing candidates with job requirements and supporting both recruiters and students.

7. CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The Résumé–Job Matching & Talent Marketplace Engine successfully demonstrates the application of Data Structures and Algorithms to a real-world recruitment problem. The system provides facilities for students to create profiles, upload résumés, and identify suitable job opportunities, while recruiters can post job requirements and find relevant candidates. Data structures and algorithms such as arrays, hashing, string matching, and priority queues are used to compare candidate skills, calculate compatibility scores, and rank candidates. The system also supports skill gap analysis, career roadmap generation, and mock interview preparation, helping students improve their employability. Overall, the project demonstrates how DSA concepts can be integrated into a practical application to improve the efficiency of candidate selection and provide better career guidance.

8.2 Future Scope

The project can be enhanced in the future with the following features:

1. Advanced Resume Parsing: Improve résumé processing to extract more detailed information such as work experience, certifications, achievements, and extracurricular activities.

2. Machine Learning-Based Matching: Integrate machine learning and natural language processing techniques to improve the accuracy of candidate-job matching.

3. Personalized Job Recommendations: Recommend suitable jobs to students based on their skills, interests, qualifications, and previous applications.

4. Advanced Skill Gap Analysis: Provide personalized learning recommendations and courses for the skills missing from a candidate's profile.

5. Interview Preparation: Expand the mock interview module with role-specific questions, automated evaluation, and performance feedback.

6. **Recruiter Analytics:** Provide recruiters with dashboards showing candidate statistics, matching scores, and recruitment trends.

7. **Scalable Database Integration:** Integrate a larger database to support more students, recruiters, job postings, and résumé records.

8. **Mobile Application:** Develop a mobile version of the platform to allow students and recruiters to access the system conveniently from smartphones.

8. REFERENCES

1. Geo, S., Alamudun, F., & Hammond, T., **RésuMatcher: A Personalized Résumé–Job Matching System,** *Expert Systems with Applications*, 2016.
https://www.sciencedirect.com/science/article/pii/S0957417416301798

2. Chala, S. A., Ansari, F., Fathi, M., & Tijdens, K., **Semantic Matching of Job Seeker to Vacancy: A Bidirectional Approach,***International Journal of Manpower*, 2018.
https://www.sciencedirect.com/science/article/pii/S0143772018000021

3. **Resume Matching Framework via Ranking and Sorting Using NLP and Deep Learning,** *2023 8th International Conference on Computer Science and Engineering (UBMK)*, 2023.
https://ieeexplore.ieee.org/abstract/document/10286605

4. **Intelligent Resume Analysis with Real-Time Multi-Platform Job Matching: A Comprehensive NLP-Driven Recruitment Automation System,** IEEE Conference Publication.
https://ieeexplore.ieee.org/abstract/document/11400433

5. **AI-Powered Resume and Job Description Matcher: A Semantic Similarity Approach Using SBERT and Contrastive Fine-Tuning,"** *2026 International Conference on Smart Futuristic Technology*.
https://ieeexplore.ieee.org/abstract/document/11507545

6. **Nexus: A Multi-Modal Framework for Semantic Job Matching and AI-Driven Resume Analysis,"** *Procedia Computer Science*, 2026.
https://www.sciencedirect.com/science/article/pii/S1877050926018041

7. Vyas, M., Chakraborty, A., & Gupta, V., **JobMatchAI – An Intelligent Job Matching Platform Using Knowledge Graphs, Semantic Search and Explainable AI,** ACL 2026.
https://aclanthology.org/2026.acl-demo.52/

8. AI-Powered Skill Gap Analysis and Career Path Recommendation System Using Hybrid Semantic and Ontology-Based Matching, IEEE Conference Publication.
https://ieeexplore.ieee.org/abstract/document/11634695